

Data Architecture

Data Architecture Specification

Gieni Acquisition Decision Intelligence Platform

Version: 1.0 | Classification: Internal Platform Architecture | Owner: Data Platform | Status: Approved



1. Purpose & Mission

The Gieni Data Platform transforms fragmented public records into acquisition-ready intelligence by modeling: Properties, Parties, Estates, Trusts, Ownership, Control, Authority, Evidence, and Outcomes.

Unlike traditional lead-generation systems that rely on flat, denormalized spreadsheets, Gieni operates through a normalized relational and graph architecture designed to support: Ownership Intelligence, Control Intelligence, Authority Resolution, Opportunity Scoring, and Outcome Learning.

2. Architectural Principles



Principle 1: Property is not opportunity.

Property ≠ Opportunity

Physical real estate is merely an underlying asset. An acquisition opportunity only exists when equity spread, statutory authority, and seller motivation converge.



Principle 2: Ownership is not control.

Ownership ≠ Control

Title records reflect historical legal vesting. De-facto control is governed by physical occupancy, family dynamics, informal leadership, or probate administration.



Principle 3: Control is not authority.

Control ≠ Authority

A motivated heir may control the physical house, but only a court-appointed fiduciary with statutory Letters possesses the legal power to execute an enforceable deed conveyance.

**Principle 4: Every conclusion must be traceable.**

All intelligence outputs must be tied to: Evidence Artifact, Schema Version, Primary Source URL, Ingestion Timestamp, and Verifying Agent / Reviewer.

**Principle 5: Every assessment is temporal.**

The system must preserve what was known, when it was known, and how it changed over time as court dockets progress from petition to closing.

3. Platform Domains

Rather than a flat cluster of tables, Gieni partitions all platform entities into six bounded operational domains:

**Domain A: Jurisdiction Domain**

- Objects: County, Court, Recorder, Assessor, Source Adapter
- Purpose: Represents municipal systems and government portals that generate primary public records.

**Domain B: Identity Domain**

- Objects: Party, Person, Organization, Contact Point, Role Assignment
- Purpose: Represents people, legal entities, and fiduciaries appearing in municipal records, establishing canonical deduplicated identity profiles.

**Domain C: Legal & Property Domain**

- Objects: Probate Case, Estate, Property, Trust, Ownership Event, Asset Interest, Stakeholder Relationship
- Purpose: Represents legal ownership structures, real estate assets, and historical deed chains.

**Domain D: Intelligence Domain**

- Objects: Ownership Assessment, Control Assessment, Authority Assessment, Valuation Assessment, Equity Assessment, Opportunity Score
- Purpose: Stores synthetic outputs, confidence ratings, and decision directives produced by intelligence engines.



Domain E: Commercial Domain

- Objects: Opportunity, Opportunity Delivery, Client, Territory Agreement, Disposition
- Purpose: Connects verified intelligence with commercial partners, governing exclusivity boundaries and delivery tracking.



Domain F: Evidence Domain

- Objects: Evidence, Claim, Claim Evidence, Review Decision, Exception, Workflow Run
- Purpose: Provides auditability, immutable SHA-256 chain-of-custody tracking, and human-in-the-loop exception management.

4. Core Entity Specifications



County Entity

- Purpose: Represents a jurisdictional territory.
- Primary Key: county_id (UUID)
- Relationships:
 - County 1:M Probate Cases
 - County 1:M Properties
 - County 1:M Territory Agreements



Probate Case Entity

- Purpose: Represents a municipal court proceeding.
- Key Fields: case_id, case_number, filing_date, case_status, case_type, presiding_judge
- Relationships:
 - Probate Case 1:1 Estate
 - Probate Case 1:M Stakeholders
 - Probate Case 1:M Authority Candidates



Property Entity

- Purpose: Represents a physical parcel reconciled against county GIS and tax rolls.
- Key Fields: property_id, apn, situs_address, legal_description, property_type, county_id
- Relationships:
 - Property M:M Asset Interests
 - Property 1:M Ownership Events
 - Property 1:M Valuation Assessments

5. Graph Relationship Model

Graph edges are explicitly separated from static entity tables. The Relationship table stores weighted, contextual graph edges between any two nodes in the system:

Relationship Edge Schema

- Schema: relationship_id (UUID), from_entity (URI), to_entity (URI), relationship_type (Enum), confidence (Float 0–1.0), source_claim (FK → Claim.claim_id)
- Relationship Types:
 - Owns / Formerly Owned
 - Controls / Potential Controller
 - Represents / Trustee Of / Beneficiary Of
 - Heir Of / Connected To / Authority Candidate
- Graph Architecture: Makes the platform graph-native, allowing graph traversal algorithms to unmask non-obvious decision pathways.

6. Intelligence Layer (Decoupled Assessments)

Gieni strictly prohibits storing intelligence conclusions inside raw source entities. Assessments are stored as discrete, versioned records linked to their respective engines:



Ownership Assessment

- Produced By: Ownership Intelligence Engine (OIE)
- Outputs: Ownership Path (Sole / Joint / TIC / Trust), Ownership Complexity Score (10–100), Ownership Confidence Rating.



Control Assessment

- Produced By: Control Intelligence Engine (CIE)
- Outputs: Control Archetype (Unified / Informal Leader / Proxy / Trust Fiduciary / Contested), Primary Controller Person ID, Consensus Requirement Flag, Control Confidence Rating.



Authority Assessment

- Produced By: Authority Resolution Engine (ARE)
- Outputs: Authority Tier (Tier 1 Certified, Tier 2 Nominated, Tier 3 Intestate Consensus, Tier 4 Unresolved), Authority Confidence Rating, Statutory Power Scope (Full Power vs Court Confirmation Required).



Opportunity Assessment

- Produced By: Opportunity Scoring Engine (OSE)
- Outputs: Composite Viability Score (0–100), Priority Band (A / B / C), Deal Friction Score (DFS), Recommended Action Playbook.

7. Evidence Architecture: Claims & Verifiable Proof



The Claim-Evidence Triad

1. Claim: Represents an atomic fact, calculation, or inference.
 - Fields: claim_id, claim_type, value, confidence, timestamp
2. Evidence: Represents immutable primary source material.
 - Examples: Probate Petition PDF, Recorded Conveyance Deed, County Assessor Roll, Court Minute Order, Trust Instrument
3. Claim Evidence: Explicitly links evidence artifacts to claims.
 - Concrete Example:
 - Claim: "John Smith is probable fiduciary"
 - Supported By: Evidence [Letters Testamentary, Page 7, Paragraph 2]
 - Hash: SHA-256 verification string

8. Opportunity Delivery Layer (Separating Intelligence from Commerce)



Opportunity vs. Opportunity Delivery

- Opportunity: Internal intelligence synthesis record (asset, authority, score, dossier).
- Opportunity Delivery: The commercial event governed by county exclusivity.
 - Fields: delivery_id, client_id, opportunity_id, delivery_date, delivery_channel (Weekly Batch / Priority A Alert), capacity_counted, billing_state
- Strategic Impact: Completely isolates proprietary intelligence from partner licensing and billing logistics.

9. The 14-Stage Opportunity Lifecycle

- | | |
|------------------------|--|
| 1. Discovered | — Court docket scraped & immutable hash assigned |
| 2. Property Identified | — Assessor GIS parcel match (PAS ≥ 70) |
| 3. Ownership Resolved | — Title deed chain & Net Equity calculated |
| 4. Control Mapped | — Informal leader & decision archetype isolated |
| 5. Authority Resolved | — Court Letters & statutory power verified |

- | | |
|-----------------|--|
| 6. Scored | — Composite Viability Score & Priority A/B/C assigned |
| 7. QC Certified | — Passed deterministic 6-gate verification |
| 8. Delivered | — Synced to exclusive partner CRM & data room |
| 9. Contacted | — Partner outreach initiated |
| 10. Appointment | — Consultation / inspection scheduled |
| 11. Offer | — Formal acquisition offer submitted |
| 12. Contract | — Purchase contract executed |
| 13. Closed | — Title recorded & wholesale / equity fee realized |
| 14. Archived | — Lifecycle completed & telemetry committed to ML training |

10. Learning Layer & Continuous Feedback Loop



The Closed-Loop Disposition Flywheel

Disposition Telemetry (Contact, Appointment, Offer, Contract, Revenue)

↓

Feedback Agent (FA) / Ingestion Webhook

↓

Model Recalibration (Adjusting County Friction & Weight Coefficients)

↓

Updated Scoring Precision & Conversion Multipliers

11. Master System Architecture Pipeline

PUBLIC RECORDS (Court Dockets & Deeds)

|

▼

PROPERTY INTELLIGENCE

|

▼

OWNERSHIP INTELLIGENCE

|

▼

CONTROL INTELLIGENCE

|

▼

AUTHORITY INTELLIGENCE

|

▼

OPPORTUNITY INTELLIGENCE

|

▼

PROBATE OPPORTUNITY FILE

|

▼

CLIENT OUTCOMES

|



Gieni converts fragmented public records into acquisition-ready decisions through a verifiable, domain-driven, graph-native intelligence platform.